10 Trigonometry

What students need to learn	Notes
A Radian measure, including use for arc length and area of sector	The formulae: $s = r\theta$ and $A = \frac{1}{2}r^2\theta$ for a circle are expected to be known.
B The three basic trigonometric ratios of angles of any magnitude (in degrees or radians) and their graphs	To include the exact values for sine, cosine and tangent of $30°, 45°, 60°$ (and the radian equivalents), and the use of these to find the trigonometric ratios of related values such as $120°, 300°$
C Applications to simple problems in two or three dimensions (including angles between a line and a plane and between two planes)	
D Use of the sine and cosine formulae	General proofs of the sine and cosine formulae will not be required. The cosine formula will be given but other formulae are expected to be known. The area of a triangle in the form: $\frac{1}{2}ab\sin C$ is expected to be known.
E The identity $\cos^2\theta + \sin^2\theta = 1$	$\cos^2\theta + \sin^2\theta = 1$ is expected to be known.
F Use of the identity $\tan\theta = \frac{\sin\theta}{\cos\theta}$	This will be provided on the formula sheet.
G The use of the basic addition formulae of trigonometry	Formal proofs of $\sin(A \pm B), \cos(A \pm B)$ formulae will not be required. Questions using the formulae for $\sin(A \pm B)$, $\cos(A \pm B)$, $\tan(A \pm B)$ may be set; the formulae will be on the formula sheet, for example: $\sin(A + B) = \sin A\cos B + \cos A\sin B$ Long questions, explicitly involving excessive manipulation, will not be set.

What students need to learn	Notes
H Solution of simple trigonometric equations for a given interval	Students should be able to solve equations such as: $\sin(x - \frac{\pi}{2}) = \frac{3}{4}$ for $0 < x < 2\pi$, $\cos(3x + 30°) = \frac{1}{2}$ for $-90° < x < 90°$, $\tan 2x = 1$ for $90° < x < 270°$, $6\cos^2 x° + \sin x° - 5 = 0$ for $0 \leqslant x < 360$, $\sin^2\left(x + \frac{\pi}{6}\right) = \frac{1}{2}$ for $-\pi \leqslant x < \pi$